

Diana He

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Mechatronics Engineering

Sept. 2025 – May 2030

3.7 GPA

TECHNICAL SKILLS

Design: SolidWorks, AutoCAD, Surface Modeling, FEA, CFD, GD&T, DFMA, Tolerance Stack-Up Analysis

Fabrication: DCIR Testing, Spot Welding, CNC Machining, Sheet Metal Fabrication, Composite Manufacturing, Resin Infusion, Wet Layup, Laser Cutting, Soldering, Mechanical-Electrical Integration, Rapid Prototyping

Programming: Python, C++, Numerical Methods

EXPERIENCE

Solar Car Battery Lead and Aerodynamics Member

Sept. 2025 – Present

Team: Midnight Sun

Promoted to Lead Aug. 2026

- Diagnosed inconsistent welds in 20+ spot-welded battery modules, fixing them through standardized sanding and adhesive application. Validated weld quality using DCIR testing, confirming resistance within ~10% of target.
- Engineered main and module-to-module busbars, using aluminum 6061 for structural rigidity and SigmaClad for high-conductivity interconnects, minimizing resistance per DFM principles.
- Redesigned module endplates and retainers to correct 6 mm of tolerance stack-up, resolving module fit issues; designed electrical mounts positioning boards near the lid for easy access and assembly within the battery box.
- Identified and redesigned a redundant wheel shield mold to consolidate it into one, reducing layup complexity and material use across 8 shields; built roof and tub rib sandwich panels via resin infusion.
- Fabricated and reinforced a fiberglass windshield mold through wet layups to withstand ~200 lb of thermoforming pressure; produced a windshield which passed competition scrutineering.

FRC Robotics Electrical Lead and Technician

Sept. 2023 – April 2025

Team: 7690 and 8349

- Engineered and implemented Team 7690's electrical wiring design within tight space constraints, prioritizing visibility and accessibility of lights and components to speed up diagnostics during competition.
- Diagnosed and repaired critical electrical and mechanical failures in high-pressure competitions, including a faulty voltage regulator, detached JST connectors, and stripped bolt holes.
- Built plexiglass clamp components using CNC and Carbide Create across 4 design revisions over 2 weeks, deburring finished parts for prototyping and testing.
- Achieved 10th/30 at Newmarket Qualifiers and 15th/50 in Science Division at Provincials.

PROJECTS

Solar Car Battery Thermal Model | *Python*

- Developed a lumped-parameter thermal model based on Midnight Sun's FSGP/ASC 2026 36S9P pack, solving 324 coupled ODEs via SciPy's implicit BDF integrator to predict peak cell temperature across a 9 hour race day.
- Derived pack current discharge from first-principles vehicle dynamics and GPS elevation data, producing a predicted SOC drawdown of 75% to 29% that closely mirrored team telemetry.
- Validated via energy-balance closure and a Biot number check; diagnosed a solver mismatch cutting closure error from 90% to under 20%, with lumping valid at nominal but marginal at worst-case conditions.
- Cross-checked against limited, partially unreliable thermistor readings averaging 50°C, attributing the gap to unmodeled auxiliary current draw, an idealized convection coefficient, and inaccurate thermistors.

Miniature Wall-E Prototype (Ongoing) | *SolidWorks, AutoCAD*

- Designed a fully articulated robotic hand in SolidWorks with 4 finger joints, using a rack-and-pinion driven sliding track for adjustable finger spacing to match WALL-E's grip.
- Derived spur gear and rack tooth geometry through analytical calculations from measured assembly clearances, resolving a 27-tooth pinion solution to fit a sub-8mm envelope within tight packaging constraints
- Optimized actuator placement by load location, using servo motors at proximal joints for lifting force and friction-fit joints distally, reducing actuator count and weight.
- Verified collision-free joint rotation and rack-pinion mesh clearance through SolidWorks interference detection and mate testing, catching geometry conflicts before committing to a print.